

Alma Mater Studiorum – Università di Bologna

Bachelor's Degree in Management and Economics

# **Backtesting and Synthetic Market Evaluation of Trading Strategies**

Supervisor:

Prof. Giovanni Cardillo

Candidate:

Jacopo Michelacci

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# Abstract

This thesis investigates the design and evaluation of systematic trading strategies using a custom backtesting framework, parameter optimization, and synthetic market generation.

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# **Chapter 1**

## **Introduction**

This chapter introduces the motivation, objectives, and scope of the thesis.

# **Chapter 2**

## **Methodology**

This chapter describes the backtesting methodology, trade accounting logic, performance metrics, and in-sample/out-of-sample evaluation protocol.

# **Chapter 3**

## **Trading Strategies**

This chapter presents the trading strategies implemented in the framework, including moving-average crossover and standard-deviation mean-reversion strategies.

# Chapter 4

## Optimization

This chapter describes the parameter optimization process and the discretionary selection of robust parameter regions to reduce overfitting risk.

# **Chapter 5**

## **Synthetic Market Generation**

This chapter presents the synthetic market generation methods used to evaluate strategy robustness under alternative market paths.



# Chapter 6

## Results

This chapter reports and discusses the empirical results obtained from historical backtests, optimization experiments, and synthetic market comparisons.

# **Chapter 7**

## **Conclusion**

This chapter summarizes the main findings, limitations, and possible future improvements.